

Statement of Purpose

I am a final-year Computer Science and Engineering student at Thapar Institute of Engineering and Technology. My interest in Machine Learning began with a fascination for systems that understand and generate human language. Whether it was a voice assistant interpreting speech, a chatbot responding to a query, or a model reasoning over large amounts of information, I was intrigued by how machines could transform raw data into meaningful understanding. This curiosity gradually led me toward Natural Language Processing, Speech Technologies, Conversational AI, and modern Transformer-based architectures.

My learning journey started with the fundamentals of machine learning, including supervised learning, neural networks, optimization, and model evaluation. As I explored deeper, I became particularly interested in language and speech. Understanding how attention mechanisms and Transformers changed the landscape of NLP motivated me to go beyond coursework and learn through research papers, technical blogs, and implementation-focused projects. Rather than treating machine learning as a collection of algorithms, I became interested in how intelligent systems are designed to understand context, recover meaning, and assist users in real-world scenarios.

One of the most impactful projects I worked on was IntentRecover AI, a context-aware semantic recovery system for dysarthric speech. Existing speech recognition systems often produce fragmented transcripts when processing impaired speech. To address this, our system first generated transcripts using Whisper and then used a T5 Transformer model to reconstruct the intended meaning from incomplete text. Working on this project exposed me to speech processing, sequence-to-sequence learning, semantic reasoning, evaluation metrics, and the challenges involved in building AI systems that improve accessibility for users with communication impairments.

More recently, I have been building RepoGraphAI, a repository intelligence platform that converts Python repositories into structured knowledge graphs using AST-based analysis. Through this project, I implemented repository parsing, graph construction, analytics, retrieval pipelines, query resolution, and context-building mechanisms for intelligent code understanding. This experience introduced me to information retrieval, knowledge graphs, graph-based reasoning, and the growing intersection between software engineering and AI-assisted systems.

I am particularly excited about Amazon ML Summer School because it offers an opportunity to learn from researchers and practitioners working on machine learning systems at real-world scale. While I have developed a strong foundation in machine learning and explored NLP, speech technologies, retrieval systems, and transformer-based models, I want to deepen my understanding of deep learning, generative AI, large language models, optimization techniques, and large-scale ML systems. Learning directly from Amazon scientists would help me bridge the gap between academic learning and industrial AI applications.

My long-term goal is to build intelligent systems that combine language, knowledge, and reasoning to solve meaningful real-world problems. I am especially interested in the intersection of NLP, information retrieval, speech technologies, and AI-powered developer tools. I believe Amazon ML Summer School will provide the technical depth, mentorship, and exposure needed to accelerate this journey, and I am eager to contribute, learn, and grow through this opportunity.